

CLASS_08: Breast Cancer Mini Project

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Background

The goal of this mini-project is for you to explore a complete analysis using the unsupervised learning techniques covered in class.

Today we will analyze a biopsy data-set from fine needle aspiration (FNA) of breast mass.

Data Import

The data is made available as a CSV file for download. We can read this in using 'read.csv()':

```
wisc.df <- read.csv("WisconsinCancer.csv", row.names = 1)
head(wisc.df, 3)
```

	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean
842302	M	17.99	10.38	122.8	1001
842517	M	20.57	17.77	132.9	1326
84300903	M	19.69	21.25	130.0	1203
	smoothness_mean	compactness_mean	concavity_mean	concave.points_mean	

842302	0.11840	0.27760	0.3001	0.14710	
842517	0.08474	0.07864	0.0869	0.07017	
84300903	0.10960	0.15990	0.1974	0.12790	
	symmetry_mean	fractal_dimension_mean	radius_se	texture_se	perimeter_se
842302	0.2419	0.07871	1.0950	0.9053	8.589
842517	0.1812	0.05667	0.5435	0.7339	3.398
84300903	0.2069	0.05999	0.7456	0.7869	4.585
	area_se	smoothness_se	compactness_se	concavity_se	concave.points_se
842302	153.40	0.006399	0.04904	0.05373	0.01587
842517	74.08	0.005225	0.01308	0.01860	0.01340
84300903	94.03	0.006150	0.04006	0.03832	0.02058
	symmetry_se	fractal_dimension_se	radius_worst	texture_worst	
842302	0.03003	0.006193	25.38	17.33	
842517	0.01389	0.003532	24.99	23.41	
84300903	0.02250	0.004571	23.57	25.53	
	perimeter_worst	area_worst	smoothness_worst	compactness_worst	
842302	184.6	2019	0.1622	0.6656	
842517	158.8	1956	0.1238	0.1866	
84300903	152.5	1709	0.1444	0.4245	
	concavity_worst	concave.points_worst	symmetry_worst		
842302	0.7119	0.2654	0.4601		
842517	0.2416	0.1860	0.2750		
84300903	0.4504	0.2430	0.3613		
	fractal_dimension_worst				
842302	0.11890				
842517	0.08902				
84300903	0.08758				

Make sure we remove the or exclude the 'diagnosis' column from the data-set that we use for further analysis- this diagnosis as either M or B

```
wisc.data <- wisc.df[,-1]
diagnosis <- as.factor( wisc.df$diagnosis )
```

Exploratory data analysis

Q1. How many observations are in this dataset?

```
nrow(wisc.data)
```

```
[1] 569
```

Q2. How many of the observations have a malignant diagnosis?

```
table(wisc.df$diagnosis)
```

```
  B   M  
357 212
```

```
sum( wisc.df$diagnosis == "M" )
```

```
[1] 212
```

Q3. How many variables/features in the data are suffixed with `_mean`?

We can use the `'grep()'` function to help us here:

```
length(grep("_mean$", colnames(wisc.data)))
```

```
[1] 10
```

Principal Component Analysis (PCA)

We need to scale our data before PCA with the `'scale=TRUE'` argument to `'prcomp()'`

```
wisc.pr <- prcomp(wisc.data, scale=TRUE)  
summary(wisc.pr)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	3.6444	2.3857	1.67867	1.40735	1.28403	1.09880	0.82172
Proportion of Variance	0.4427	0.1897	0.09393	0.06602	0.05496	0.04025	0.02251
Cumulative Proportion	0.4427	0.6324	0.72636	0.79239	0.84734	0.88759	0.91010
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
Standard deviation	0.69037	0.6457	0.59219	0.5421	0.51104	0.49128	0.39624
Proportion of Variance	0.01589	0.0139	0.01169	0.0098	0.00871	0.00805	0.00523
Cumulative Proportion	0.92598	0.9399	0.95157	0.9614	0.97007	0.97812	0.98335
	PC15	PC16	PC17	PC18	PC19	PC20	PC21
Standard deviation	0.30681	0.28260	0.24372	0.22939	0.22244	0.17652	0.1731
Proportion of Variance	0.00314	0.00266	0.00198	0.00175	0.00165	0.00104	0.0010
Cumulative Proportion	0.98649	0.98915	0.99113	0.99288	0.99453	0.99557	0.9966

	PC22	PC23	PC24	PC25	PC26	PC27	PC28
Standard deviation	0.16565	0.15602	0.1344	0.12442	0.09043	0.08307	0.03987
Proportion of Variance	0.00091	0.00081	0.0006	0.00052	0.00027	0.00023	0.00005
Cumulative Proportion	0.99749	0.99830	0.9989	0.99942	0.99969	0.99992	0.99997
	PC29	PC30					
Standard deviation	0.02736	0.01153					
Proportion of Variance	0.00002	0.00000					
Cumulative Proportion	1.00000	1.00000					

Q4. From your results, what proportion of the original variance is captured by the first principal component (PC1)?

Proportion of Variance for PC1: $0.4427 = 44.27\%$

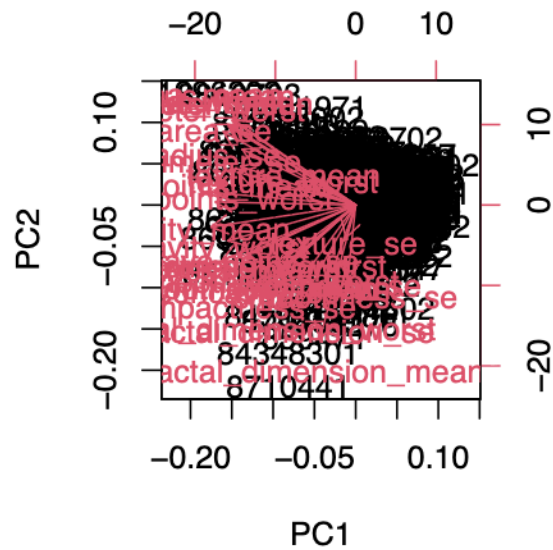
Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

Cumulative Proportion and find first value 0.70: $PC1 = 0.4427$ $PC2 = 0.6324$ $PC3 = 0.72636$ -
therefore, 3 PCs

Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

$PC6 = 0.88759$ $PC7 = 0.91010$ Therefore, 7 PCs

```
biplot(wisc.pr)
```



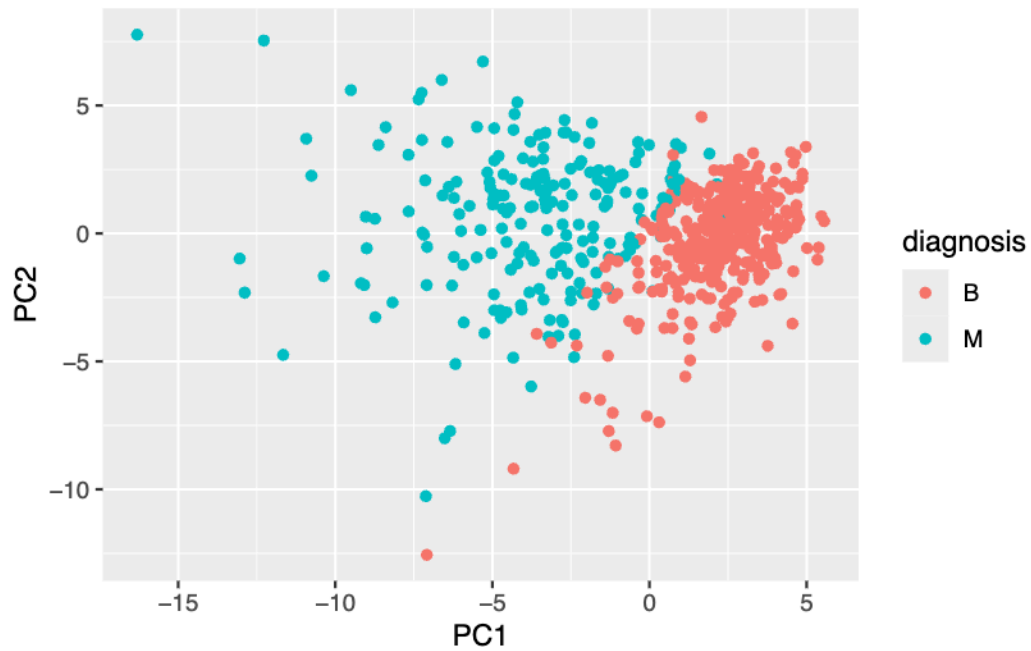
Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

The biplot is difficult to interpret because the many sample labels and variable arrows overlap heavily, making the graph cluttered. It is useful for showing that PCA captures structure in the data, but clearer custom plots are needed to interpret the results well.

Let's see the "PC score plot"

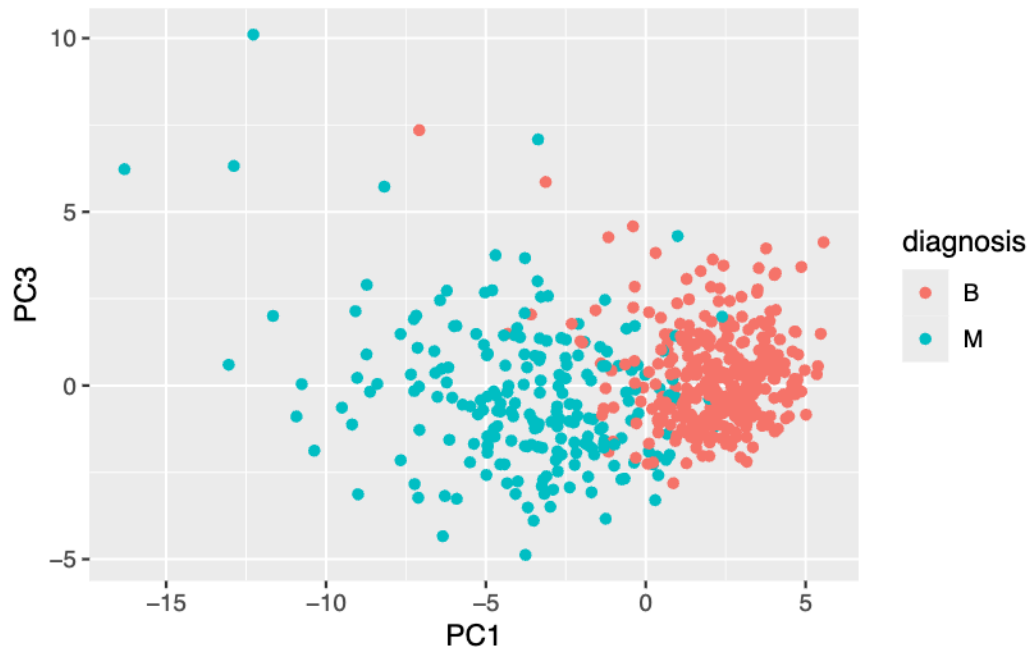
```
library(ggplot2)

ggplot(wisc.pr$ x) +
  aes(PC1, PC2, col=diagnosis) +
  geom_point()
```



Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

```
ggplot(wisc.pr$x) +  
  aes(PC1, PC3, col=diagnosis) +  
  geom_point()
```



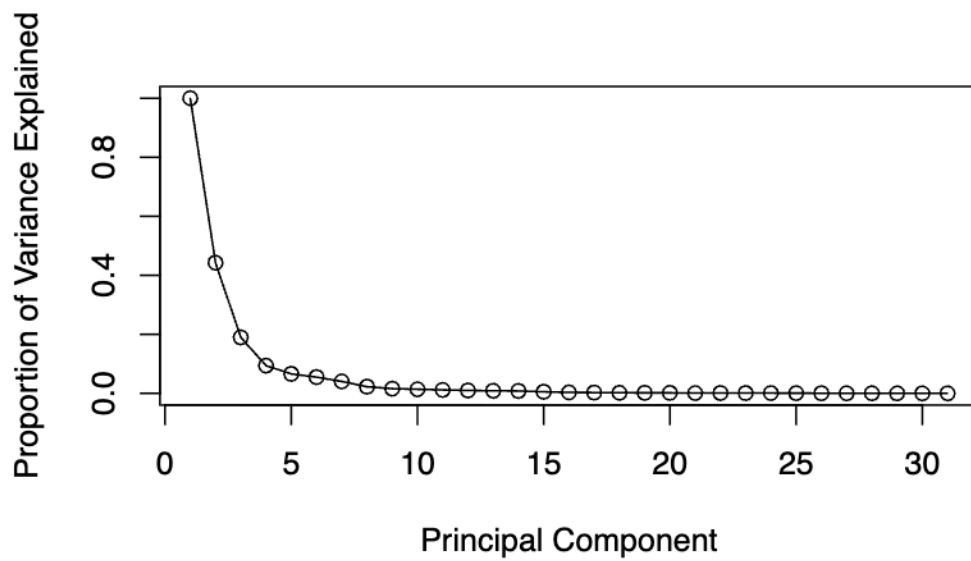
PC1 still shows the clearest separation between benign and malignant samples, while PC3 adds much less additional separation. This suggests most of the diagnostic signal is captured by PC1, and later components contribute less to distinguishing the two groups.

```
# Calculate variance of each component
pr.var <- wisc.pr$sdev^2
head(pr.var)
```

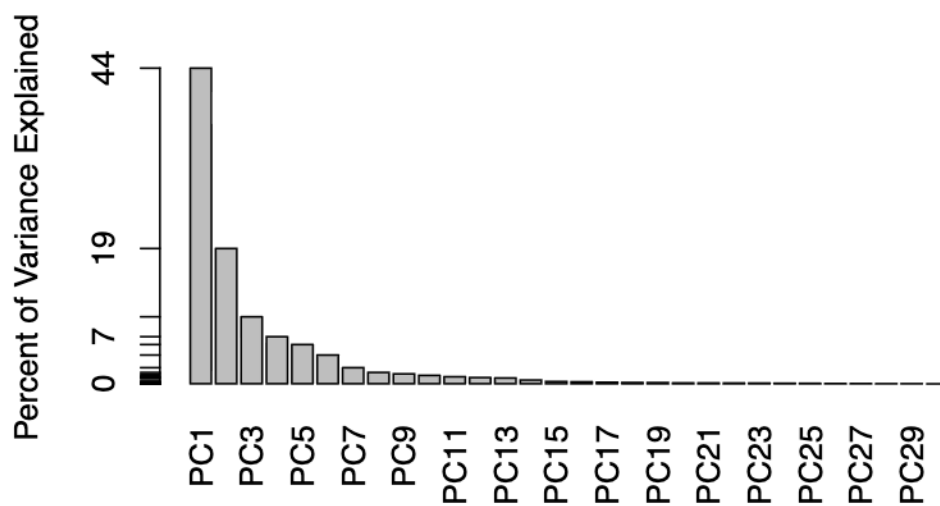
```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

```
# Variance explained by each principal component
pve <- pr.var / sum(pr.var)

# Plot variance explained
plot(c(1,pve), xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     ylim = c(0,1), type = "o")
```



```
barplot(pve, ylab = "Percent of Variance Explained",
        names.arg = paste0("PC", 1:length(pve)), las = 2, axes = FALSE)
axis(2, at = pve, labels = round(pve, 2) * 100)
```



Q9. For the first principal component, what is the component of the loading vector (i.e. `wisc.pr$rotation[,1]`) for the feature `concave.points_mean`? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

```
wisc.pr$rotation["concave.points_mean",1]
```

```
[1] -0.2608538
```

```
sort(abs(wisc.pr$rotation[,1]), decreasing=TRUE)
```

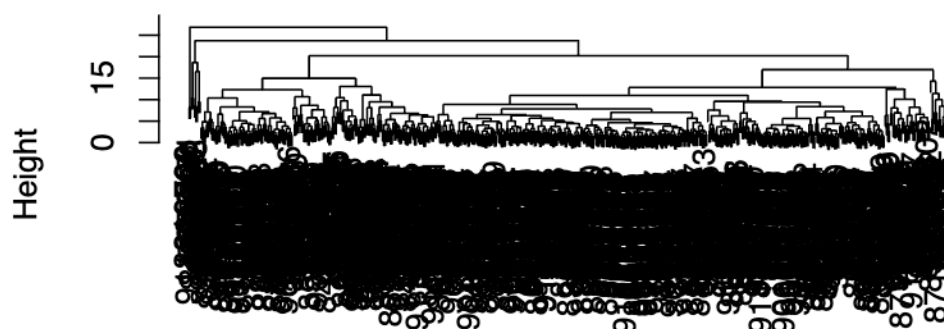
concave.points_mean	concavity_mean	concave.points_worst
0.26085376	0.25840048	0.25088597
compactness_mean	perimeter_worst	concavity_worst
0.23928535	0.23663968	0.22876753
radius_worst	perimeter_mean	area_worst
0.22799663	0.22753729	0.22487053
area_mean	radius_mean	perimeter_se
0.22099499	0.21890244	0.21132592
compactness_worst	radius_se	area_se
0.21009588	0.20597878	0.20286964
concave.points_se	compactness_se	concavity_se
0.18341740	0.17039345	0.15358979
smoothness_mean	symmetry_mean	fractal_dimension_worst
0.14258969	0.13816696	0.13178394
smoothness_worst	symmetry_worst	texture_worst
0.12795256	0.12290456	0.10446933
texture_mean	fractal_dimension_se	fractal_dimension_mean
0.10372458	0.10256832	0.06436335
symmetry_se	texture_se	smoothness_se
0.04249842	0.01742803	0.01453145

The loading for `concave.points_mean` on PC1 is -0.2609 (magnitude 0.2609). No, there are no larger contributions than this one, so it is the strongest contributor to PC1.

Hierarchical clustering

```
wisc.hclust <- hclust(dist( scale(wisc.data) ) )
plot(wisc.hclust)
```

Cluster Dendrogram

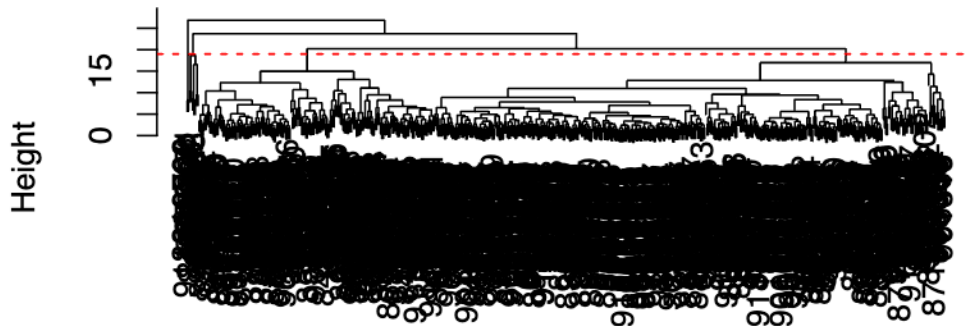


```
dist(scale(wisc.data))  
hclust (*, "complete")
```

Q10. Using the `plot()` and `abline()` functions, what is the height at which the clustering model has 4 clusters?

```
plot(wisc.hclust)  
abline(h=19, col="red", lty=2)
```

Cluster Dendrogram



```
dist(scale(wisc.data))  
hclust(*, "complete")
```

The clustering model has 4 clusters at approximately height = 19.

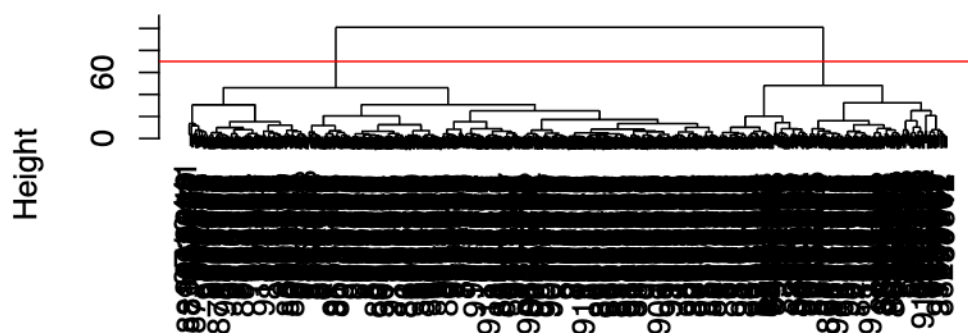
Q12. Which method gives your favorite results for the same data.dist dataset?
Explain your reasoning.

ward.D2 gives the best results because it minimizes within-cluster variance and forms compact, balanced clusters. It usually produces clearer separation and less chaining than single, complete, or average linkage for this dataset.

Combining methods

```
d <- dist(wisc.pr$x[,1:4])  
wisc.pr.hclust <- hclust(d, method="ward.D2")  
plot(wisc.pr.hclust)  
abline(h=70, col="red")
```

Cluster Dendrogram



d
hclust (*, "ward.D2")

```
grps <- cutree(wisc.pr.hclust, h=70)
table(grps)
```

```
grps
  1  2
171 398
```

How does this clustering 'grps' correspond to the expert 'diagnosis' ?

```
table(diagnosis)
```

```
diagnosis
  B  M
357 212
```

```
table(diagnosis, grps)
```

```
      grps
diagnosis  1  2
  B      6 351
  M     165  47
```

Q13. How well does the newly created hclust model with two clusters separate out the two “M” and “B” diagnoses?

```
wisc.pr.hclust.clusters <- cutree(wisc.pr.hclust, k = 2)
table(wisc.pr.hclust.clusters, diagnosis)
```

	diagnosis	
wisc.pr.hclust.clusters	B	M
1	6	165
2	351	47

Q14. How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses? Again, use the table() function to compare the output of each model (wisc.hclust.clusters and wisc.pr.hclust.clusters) with the vector containing the actual diagnoses.

```
wisc.hclust.clusters <- cutree(wisc.hclust, k = 4)
table(wisc.hclust.clusters, diagnosis)
```

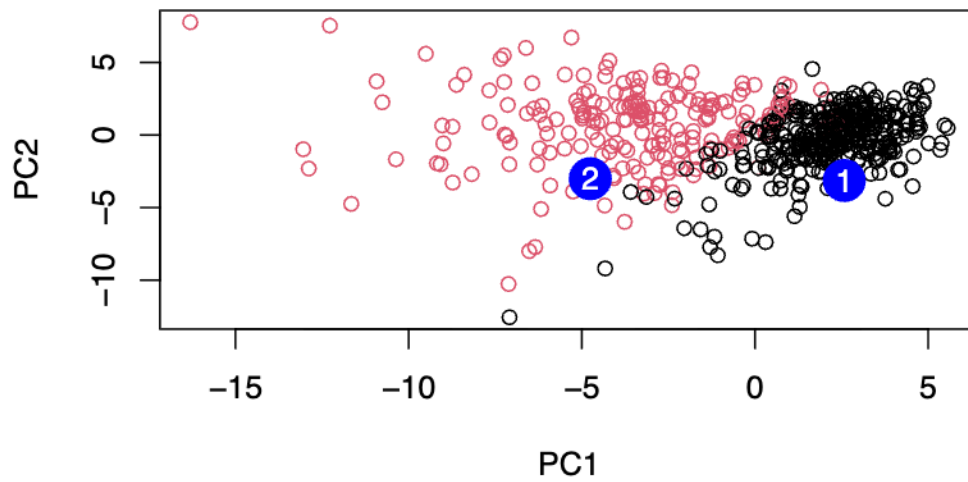
	diagnosis	
wisc.hclust.clusters	B	M
1	12	165
2	2	5
3	343	40
4	0	2

Prediction

```
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)
npc <- predict(wisc.pr, newdata=new)
```

```
grps <- as.factor(diagnosis)
```

```
plot(wisc.pr$x[,1:2], col=grps)
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)
text(npc[,1], npc[,2], c(1,2), col="white")
```



Q16. Which of these new patients should we prioritize for follow up based on your results?

Patient 2 should be prioritized for follow-up because it falls within the cluster associated with malignant tumors, while Patient 1 is grouped with the benign cluster. This suggests Patient 2 has a higher likelihood of malignancy and needs faster clinical evaluation.